

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

Task to be Done

1. The Catalog Search (JOINS): Create a third table called Loans that connects member_id to book_id. Write a query to show the Member Name and the Book Title they have currently borrowed.
2. Collection Stats (Aggregations): Write a query to find the total number of books published in each year. o Hint: Use COUNT(book_id) and GROUP BY published_year.
3. The Donation Transaction (ACID): Use a Transaction (BEGIN...COMMIT) to add a new book to the Books table and simultaneously log the entry in a Donation_History table.
4. Search Speed (Indexing): Create an Index on the isbn column. Explain why searching by ISBN is now faster for the librarian than searching by Title.

Skill Set Gained

- Relational Logic: Connecting independent tables (Books + Members) through a Junction table (Loans).
- Analytical SQL: Using GROUP BY to generate business insights.
- Atomic Operations: Using Transactions to ensure multiple database changes succeed or fail as a single unit.

```

mysql> USE bookflow_db;
Database changed
mysql>
mysql> CREATE TABLE Loans (
->     loan_id INT PRIMARY KEY,
->     member_id INT,
->     book_id INT,
->     loan_date DATE,
->     return_date DATE,
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
ERROR 1050 (42S01): Table 'loans' already exists
mysql>
mysql>
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date, return_date)
-> VALUES
-> (1, 1, 2, '2026-08-10', NULL),
-> (2, 2, 1, '2026-08-12', NULL),
-> (3, 3, 3, '2026-08-14', '2026-08-16');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT
->     Members.full_name AS Member_Name,
->     Books.title AS Book_Title
-> FROM Loans
-> JOIN Members
->     ON Loans.member_id = Members.member_id
-> JOIN Books
->     ON Loans.book_id = Books.book_id
-> WHERE Loans.return_date IS NULL;
+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| Rahul Kumar | Atomic Habits |
| Priya Sharma | The Alchemist |
+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT
->     published_year,
->     COUNT(book_id) AS total_books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
+-----+-----+
| published_year | total_books |
+-----+-----+
| 1988 | 1 |
| 2008 | 1 |
| 2018 | 1 |
+-----+-----+
3 rows in set (0.01 sec)

```

```
mysql> CREATE TABLE Donation_History (  
->     donation_id INT PRIMARY KEY,  
->     book_id INT,  
->     donation_date DATE,  
->     donor_name VARCHAR(100),  
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );  
Query OK, 0 rows affected (0.06 sec)  
  
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql>  
mysql> INSERT INTO Books (book_id, title, isbn, published_year)  
-> VALUES (4, 'The Pragmatic Programmer', '9780135957059', 1999);  
Query OK, 1 row affected (0.00 sec)  
  
mysql>  
mysql> INSERT INTO Donation_History  
->     (donation_id, book_id, donation_date, donor_name)  
-> VALUES  
->     (1, 4, '2026-08-17', 'Ravi Kumar');  
Query OK, 1 row affected (0.01 sec)  
  
mysql>  
mysql> COMMIT;  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> ROLLBACK;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql>  
mysql> INSERT INTO Books (book_id, title, isbn, published_year)  
-> VALUES (5, 'Database Systems', '9781234567890', 2020);  
Query OK, 1 row affected (0.00 sec)  
  
mysql>  
mysql> INSERT INTO Donation_History  
->     (donation_id, book_id, donation_date, donor_name)  
-> VALUES  
->     (2, 5, '2026-08-17', 'Anil Kumar');  
Query OK, 1 row affected (0.00 sec)  
  
mysql>  
mysql> COMMIT;  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> ROLLBACK;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql>  
mysql> CREATE INDEX idx_books_isbn
```

```
-> (donation_id, book_id, donation_date, donor_name)
-> VALUES
-> (2, 5, '2026-08-17', 'Anil Kumar');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> CREATE INDEX idx_books_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SHOW INDEX FROM Books;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| books | 0 | PRIMARY | 1 | book_id | A | 5 | NULL | NULL | NULL | YES | BTREE | | YES | NULL |
| books | 1 | idx_books_isbn | 1 | isbn | A | 5 | NULL | NULL | NULL | YES | BTREE | | YES | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)

mysql>
mysql> SELECT *
-> FROM Books
-> WHERE isbn = '9780135957059';
+-----+-----+-----+-----+
| book_id | title | isbn | published_year |
+-----+-----+-----+-----+
| 4 | The Pragmatic Programmer | 9780135957059 | 1999 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```